

Unsupervised Learning and Dimensionality Reduction: Clustering, Projection, and Neural Network Analysis

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Abstract—This report investigates unsupervised learning through clustering and dimensionality reduction on two benchmark datasets: Adult Income and Wine Quality. K-Means and EM/GMM clustering are applied alongside PCA, ICA, Randomized Projections, and UMAP, evaluating how transformed feature spaces affect cluster structure and neural network performance. PCA retains approximately 72% of Wine Quality variance in four components and achieves the strongest cluster separation across both datasets, raising Wine silhouette from 0.247 to 0.447. Dimensionality reduction to four components reduces Wine classification accuracy from 65.0% to 60.0–64.0%, trading modest predictive performance for faster convergence. Augmenting the original features with cluster-derived encodings recovers near-perfect training accuracy at the cost of overfitting, underscoring that cluster features supplement but do not replace raw physicochemical measurements.

I. INTRODUCTION

Unsupervised learning seeks to expose intrinsic structure in data without labeled targets. Clustering algorithms such as K-Means and EM/GMM partition observations into groups sharing distributional similarity, while dimensionality reduction methods such as PCA, ICA, and random projection find compact representations that preserve dominant structure in the original feature space. The two families are deeply inter-related: high-dimensional spaces are hostile to distance-based clustering as pairwise distances concentrate and lose discriminative power [5], and prior projection can restore the geometric contrast that clustering relies on. Cluster assignments, in turn, can serve as engineered features for downstream supervised models.

This report applies both families to Adult Income and Wine Quality, extending prior supervised and online learning experiments to enable direct comparison across the full pipeline. Three questions guide the analysis: does dimensionality reduction improve cluster quality, and which method works best; does compressing the feature space trade accuracy for training speed; and do cluster-derived features help or hurt a downstream classifier? The two datasets provide a useful contrast—Adult is a sparse, high-dimensional binary space while Wine is dense and continuously valued—so the two regimes illuminate different failure modes for the same algorithms.

II. DATASETS

A. Dataset A: Adult Income

The Adult Income (Census Income) dataset [1] contains 48,842 instances drawn from the 1994 U.S. Census, with

14 attributes spanning demographic and employment characteristics. The binary target encodes whether annual income exceeds \$50K. Raw categorical features—workclass, education, marital status, occupation, relationship, race, sex, and native country—were one-hot encoded, yielding 108 binary columns. Continuous features (age, fnlwtg, education-num, capital-gain, capital-loss, hours-per-week) were standardized with `StandardScaler`; missing values encoded as “?” were replaced with the modal category per column.

The dataset is class-imbalanced ($\approx 75\%$ negative, 25% positive). Capital-gain and capital-loss are extremely right-skewed, which is why `StandardScaler` was preferred over `MinMaxScaler`—the latter would let a handful of large outliers dominate the scale. The resulting 108-feature binary space is a challenging regime for distance-based methods, since Euclidean distance loses discriminative power in sparse spaces [5], making dimensionality reduction a natural first step before clustering.

B. Dataset B: Wine Quality

The Wine Quality dataset [2] combines red and white Portuguese Vinho Verde samples ($n = 6,497$; 1,599 red, 4,898 white) with 11 physicochemical input features and a discrete quality rating from 3 to 9, treated as a seven-class target. Features were scaled to $[0, 1]$ with `MinMaxScaler` to reduce magnitude differences across attributes such as pH (≈ 2.7 – 4.0) and total sulfur dioxide (≈ 6 – 289 mg/L). A binary color indicator was appended as a 12th feature.

Classes 5 and 6 together account for $\approx 70\%$ of samples, while classes 3, 4, 8, and 9 together represent less than 5% . This severe imbalance sets a ceiling on how well any unsupervised algorithm can recover quality labels—there simply are not enough examples of the extreme classes to form stable clusters. Strong inter-feature correlations further motivate PCA and ICA: free and total sulfur dioxide ($r \approx 0.72$), density and alcohol ($r \approx -0.69$), and volatile acidity and quality ($r \approx -0.27$). These relationships underpin the hypotheses stated below.

C. Hypotheses

H1 (PCA Compactness): PCA will capture over 70% of Wine Quality variance in four or fewer components because physicochemical attributes such as acidity, sulfur dioxide, and alcohol are strongly correlated within wine type.

H2 (ICA Independence): ICA will isolate at least one high-kurtosis component corresponding to a meaningful chemical dimension (e.g., the alcohol-density fermentation axis) that is obscured in the PCA basis.

H3 (Cluster-Label Alignment): K-Means and EM will not cleanly recover the seven Wine quality classes from raw features, but dimensionality reduction—particularly PCA—will improve cluster separation as measured by silhouette score.

III. STEP 1: CLUSTERING ON RAW DATA

A. Algorithm Overview

K-Means partitions n observations into k clusters by iteratively minimizing the within-cluster sum of squared Euclidean distances [3]. It assumes convex, isotropic clusters of approximately equal size and is sensitive to initialization, so we used k -means++ seeding [6] with 20 independent restarts, retaining the solution with the lowest inertia. Convergence was defined as a relative change in inertia below 10^{-4} .

EM for Gaussian Mixture Models (GMM) generalizes K-Means by treating cluster membership as a latent variable and fitting full covariance ellipsoids via alternating E and M steps [7]. The E-step computes soft responsibilities $\gamma_{ik} = p(z_i = k | x_i)$; the M-step updates means, covariances, and mixing weights. GMM can model elliptical and overlapping clusters but is more susceptible to numerical singularities when covariances become ill-conditioned—hence the regularization term $\text{reg_covar} = 10^{-6}$ and 5 random initializations used here. For both algorithms, k was swept across $\{2, 3, 4, 5, 6, 7, 8, 10, 12, 15\}$, with final configurations selected using silhouette, Calinski-Harabasz (CH), Davies-Bouldin (DB), and BIC (GMM only). ARI is reported post-hoc as an interpretive reference only, not as a selection criterion.

B. Wine Quality Clustering

On the 12-dimensional scaled Wine feature space, both algorithms struggle to recover fine-grained quality ratings. For K-Means, the silhouette score peaks at $k = 2$ (silhouette ≈ 0.35), which turns out to correspond to a coarse red-vs-white split driven by the color indicator and correlated sulfur dioxide ranges rather than anything quality-related. The CH index also peaks at $k = 2$ but shows a secondary elbow at $k = 4$, and the DB index is relatively flat between $k = 3$ and $k = 5$. Given that $k = 2$ collapses to a wine-type partition with no quality signal, $k = 4$ was selected as a compromise: two clusters align approximately with wine type, while the remaining two begin to differentiate higher- from lower-quality samples within each type. Table I summarizes the metrics at $k = 4$.

TABLE I
CLUSTERING ON WINE QUALITY (RAW FEATURES, $k = 4$)

Algorithm	Silhouette	CH Index	DB Index	ARI
K-Means	0.247	1921.5	1.536	0.150
EM (GMM)	0.230	1726.7	1.816	0.162

The low ARI values confirm H3: quality labels are not cleanly recoverable from raw-feature clustering. Each cluster contains a mix of ratings centered around 5 and 6, with only weak enrichment for extreme ratings in any single cluster—an outcome that is expected given the class imbalance noted in Section II. The GMM edges out K-Means on ARI (0.162 vs. 0.150), plausibly because its flexible covariance better captures the overlapping quality-tier geometry that K-Means’ isotropic assumption cannot model, though the difference is modest enough that it should not be over-interpreted.

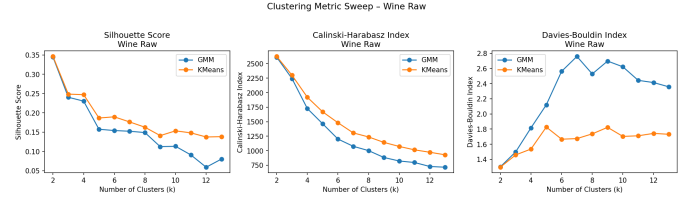


Fig. 1. Clustering metric sweep on Wine Quality raw features. The $k = 2$ peak in silhouette reflects the wine-type split, not quality structure—selecting it would recover red vs. white, not what this analysis targets. The secondary elbow at $k = 4$ in CH, combined with the flat DB between $k = 3$ and $k = 5$, is why $k = 4$ was chosen: it is the smallest k that begins to differentiate quality tiers beyond the type partition.

C. Adult Income Clustering

The 108-dimensional one-hot encoded Adult space presents a sparse binary feature landscape. Ideally one would use Hamming or Jaccard distance for binary data, but Euclidean distance was retained for methodological consistency across all experiments; this limitation is worth keeping in mind when interpreting the results. As expected [5], silhouette scores are uniformly low across all tested k (peak ≈ 0.13 near $k = 4$), and the inertia curve decreases monotonically with no clear elbow—both classic symptoms of the curse of dimensionality, where all pairwise distances converge toward similar values and the notion of a nearest centroid becomes unreliable.

Despite this, K-Means at $k = 4$ recovers ARI of 0.095, suggesting at least a weak alignment with the income label. GMM at $k = 4$ yields higher ARI (0.126) but lower silhouette (0.081). The $k = 4$ selection here is admittedly less clean than for Wine—the silhouette peak is shallow and the CH curve gives limited guidance—but it represents the best available trade-off given the absence of a sharper signal in the metric sweeps, and it keeps the cluster count comparable across datasets for easier comparison. Table II reports the full metrics.

TABLE II
CLUSTERING ON ADULT INCOME (RAW FEATURES, $k = 4$)

Algorithm	Silhouette	CH Index	DB Index	ARI
K-Means	0.130	5659.4	1.710	0.095
EM (GMM)	0.081	2601.2	3.054	0.126

The CH index strongly favors K-Means (5659.4 vs. 2601.2) because it rewards tight, well-separated hard assignments—a regime where GMM’s soft boundaries are a disadvantage. The

divergent metric signals here reinforce the point that no single configuration dominates in high-dimensional binary spaces, and any choice of k involves accepting some ambiguity.

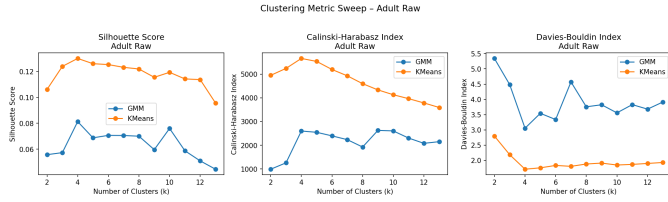


Fig. 2. Clustering metric sweep on Adult Income raw features. The absence of a clear elbow in the inertia curve and the uniformly low, flat silhouette scores across all k are the operational signature of the curse of dimensionality: the 108-dimensional one-hot space offers no compact cluster structure for either algorithm to latch onto, which motivates dimensionality reduction in Step 3.

IV. STEP 2: DIMENSIONALITY REDUCTION

A. PCA

PCA was applied after `StandardScaler` for Wine and after one-hot encoding plus `StandardScaler` for Adult. For Wine, the full 12-component PCA was first computed to inspect the eigenvalue spectrum. The cumulative explained variance curve exhibits a pronounced elbow: components 1 through 4 account for approximately 32%, 52%, 65%, and 72% of total variance respectively, supporting H1. Components 5–12 each contribute less than 4% individually, which means that most of the useful linear structure in the Wine features really does live in four dimensions.

The loading vectors tell a coherent chemical story. PC1 loads most heavily on free sulfur dioxide, total sulfur dioxide, and the wine color indicator—sulfite chemistry and wine type dominate the primary axis. PC2 loads on volatile and fixed acidity, capturing the acidity profile that separates red from white. PC3 reflects the residual sugar–density relationship tied to fermentation level, and PC4 picks up alcohol and pH, which track fermentation endpoint. The fact that these four components map onto recognizable chemical processes rather than arbitrary directions is a useful sanity check that PCA is capturing real structure and not just noise.

For Adult Income, the eigenvalue spectrum is considerably flatter: the first 20 principal components capture approximately 89% of variance, and there is no clear elbow anywhere in the curve. This is a natural consequence of the one-hot encoding—each binary indicator contributes a small, roughly equal share of variance, so PCA has no dominant direction to concentrate onto. We select 20 components as a practical threshold that retains $\approx 89\%$ of variance while reducing dimensionality by 81%, though any nearby choice would have been similarly defensible given the absence of a sharp elbow.

B. ICA

FastICA [4] was applied with PCA whitening as a mandatory preprocessing step, which decorrelates the data and normalizes component variances before the fixed-point optimization for non-Gaussianity. For Wine, four independent

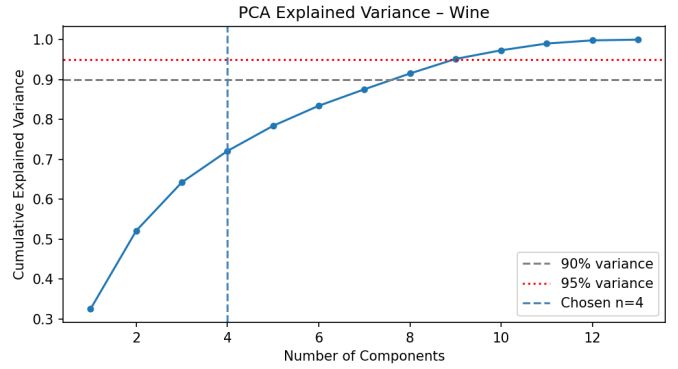


Fig. 3. PCA cumulative explained variance for Wine Quality. The sharp elbow after component 4 (dashed blue, $\approx 72\%$) is what makes $n = 4$ a principled choice rather than an arbitrary one—it corresponds to the point where adding another component yields diminishing returns in variance explained, and maps onto a physically interpretable four-axis chemical space.

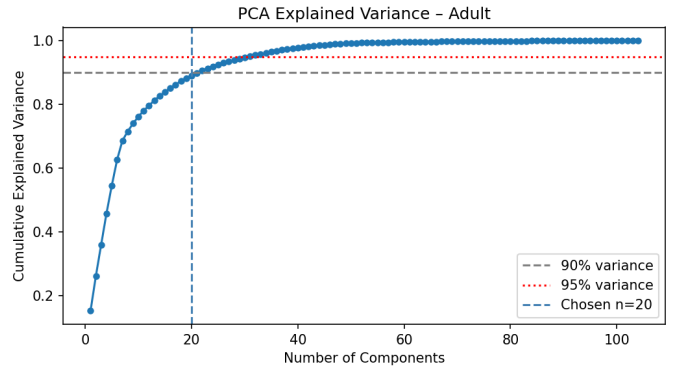


Fig. 4. PCA cumulative explained variance for Adult Income. The lack of an elbow in the spectrum (chosen $n = 20$, dashed blue) is itself informative: it means PCA cannot compress Adult the way it compresses Wine, and any reduced representation will lose a meaningful fraction of the one-hot structure. This diffuseness predicts that PCA will offer less clustering benefit for Adult than for Wine, a prediction tested in Step 3.

components were extracted using the logistic contrast function $g(u) = \tanh(u)$, convergent within 200 iterations for all restarts.

Absolute excess kurtosis (Fisher definition) was computed for each component as a proxy for non-Gaussianity. The Wine components yield values of 1.29, 0.06, 0.97, and 3.86. Component 4 stands out clearly at $\kappa \approx 3.86$, supporting H2: inspecting the unmixing matrix, it loads most strongly on alcohol and density with opposite signs, which maps directly onto the fermentation biochemistry in which higher alcohol corresponds to lower density as sugar is converted. This is a single, interpretable independent source that PCA distributes across PC3 and PC4. The two near-Gaussian components ($\kappa \approx 0.06$ and 0.97) are less useful and, as will be seen in Step 3, they limit ICA’s clustering benefit relative to PCA.

For Adult, 20 ICA components were extracted. Most kurtosis values fall below 6, but component 15 reaches ≈ 150 and component 14 reaches approximately 19. These extreme values correspond to rare binary indicator variables—specific

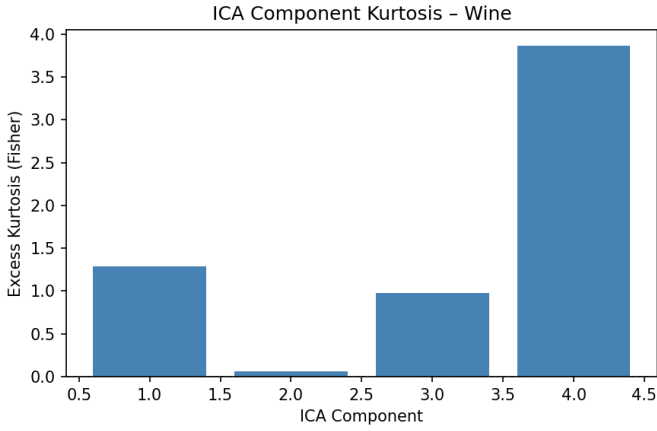


Fig. 5. ICA component excess kurtosis for Wine Quality (4 components). Component 4’s dominant kurtosis ($\kappa \approx 3.86$) isolates the alcohol-density fermentation axis as a single direction, a structure that PCA smears across two components. The near-zero kurtosis of component 2 ($\kappa \approx 0.06$) signals that this direction carries no independent chemical signal and will contribute noise rather than structure in downstream clustering.

native-country or occupation categories with very few positive examples—that produce sharply spiked marginal distributions. Technically this satisfies the ICA non-Gaussianity criterion, but it does not represent a meaningful independent source. The practical consequence for clustering is examined in Step 3.

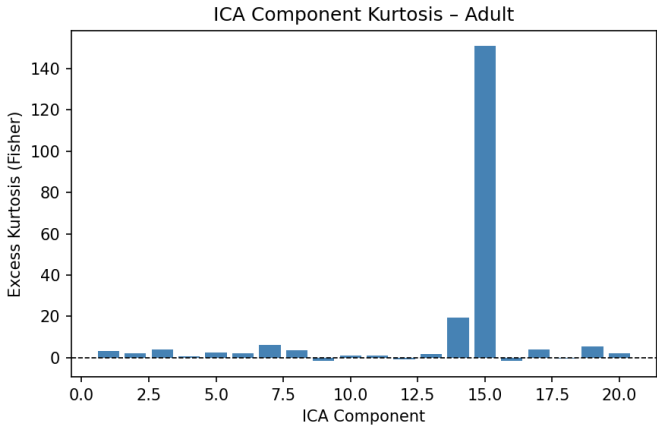


Fig. 6. ICA component excess kurtosis for Adult Income (20 components). The extreme spike at component 15 ($\kappa \approx 150$) is not a meaningful independent chemical or demographic source—it is a categorical co-occurrence artifact from a rare binary indicator. Its presence in the basis distorts clustering in Step 3 by creating an artificial high-density micro-region that pulls cluster centers away from the income-relevant structure.

C. Randomized Projection

Gaussian Random Projection (GRP) [9] was applied by sampling the projection matrix $R \in \mathbb{R}^{d_{in} \times d_{out}}$ with entries $r_{ij} \sim \mathcal{N}(0, 1/d_{out})$. Reconstruction error was estimated via the Moore-Penrose pseudoinverse: $\hat{X} = XR(R^T R)^{-1}R^T$, with MSE averaged over all samples and features across five

independent random seeds to capture both mean error and seed-to-seed variance.

For Wine at $d = 4$, mean reconstruction MSE ≈ 0.61 with seed standard deviation of 0.047. This is substantially higher than what PCA achieves at the same dimensionality—an expected result, since RP preserves pairwise distances in expectation but does not concentrate variance into dominant directions, so a random 4-dimensional projection discards more reconstructable information than PCA’s optimal one. We select $d_{out} = 4$ for Wine to enable direct comparison with PCA and ICA, and note that the moderate seed variance means individual RP runs will produce somewhat different projections, which is worth keeping in mind when interpreting downstream results.

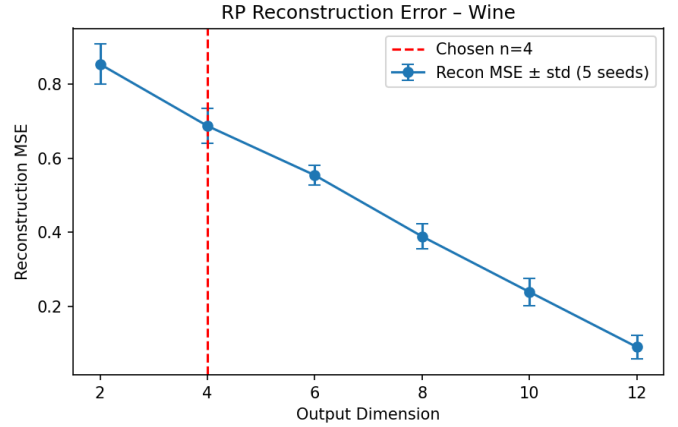


Fig. 7. RP reconstruction MSE vs. output dimension for Wine Quality (mean $\pm$ std over 5 seeds). The chosen $n = 4$ (dashed red) sits on a steeply declining curve, meaning the projection at this dimensionality is quite lossy. That RP still performs competitively in the Step 4 neural network despite this high reconstruction error reveals something important: the network’s decision boundary depends on relative sample geometry, which RP preserves, not on absolute reconstructability, which it does not.

For Adult at $d = 20$, mean MSE ≈ 0.107 with near-zero seed variance (std ≈ 0.0015), roughly comparable to PCA at the same dimensionality. The parity makes sense: since Adult’s variance is diffusely spread across 108 dimensions, a random projection to 20 dimensions loses roughly the same fraction of variance as the optimal PCA projection, because neither method can concentrate the diffuse spectrum. The low seed variance—much lower than Wine—reflects the law of large numbers operating over a larger input space.

D. UMAP (Extra Credit)

UMAP [10] was applied with `n_neighbors=15` and `min_dist=0.1` to produce two-dimensional embeddings for visualization only. These embeddings were not used as classifier inputs, consistent with the course guidance on UMAP’s role as an analysis tool.

For Wine, the 2-D embedding reveals a clear bifurcation corresponding to red and white wines, forming two dense, well-separated manifolds with minimal overlap. Within each manifold, quality gradients appear as smooth color transitions,

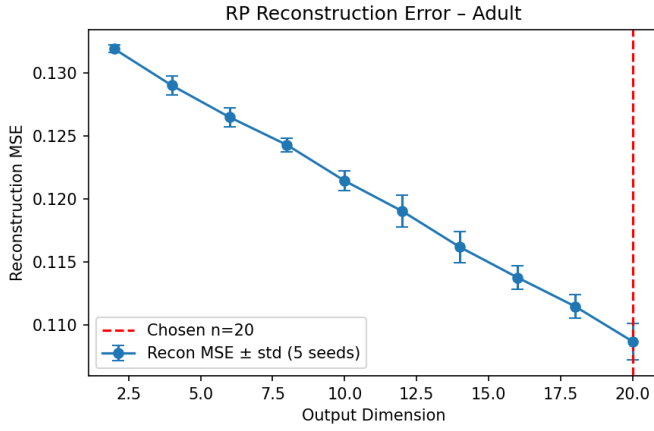


Fig. 8. RP reconstruction MSE vs. output dimension for Adult Income (mean $\pm$ std over 5 seeds). The near-zero seed variance at $n = 20$ (dashed red) contrasts with Wine’s higher variance and reflects the large input dimensionality stabilizing individual projections. The absence of a plateau in the error curve mirrors the flat PCA eigenspectrum: Adult has no compact variance structure for either method to exploit, so error decreases gradually rather than leveling off.

suggesting that wine quality lies on a low-dimensional manifold that UMAP successfully unrolls—a topology that the first two PCA components obscure because the linear projection overlaps the two wine types. This confirms that meaningful non-linear structure exists in the Wine feature space, beyond what linear methods expose.

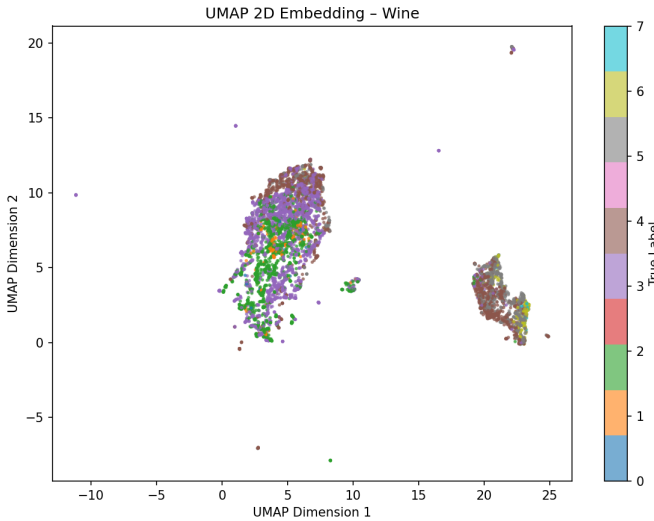


Fig. 9. UMAP 2-D embedding for Wine Quality colored by true quality label. The two well-separated manifolds correspond to the red/white wine split, which is the same dominant structure found by PCA and K-Means at $k = 2$. The smoother quality gradients visible within each manifold are what linear methods miss, and they suggest that additional quality-tier separation might be achievable with non-linear methods applied to a larger or cleaner dataset.

For Adult, UMAP reveals dense regions corresponding to specific occupation-education-marital-status combinations, reflecting categorical co-occurrence structure. The binary income boundary, however, is diffuse throughout the embedding

with positive and negative class members interleaved in most regions—consistent with the low ARI values in Step 1 and suggesting that the income boundary is not well-described by any low-dimensional manifold of these features alone.

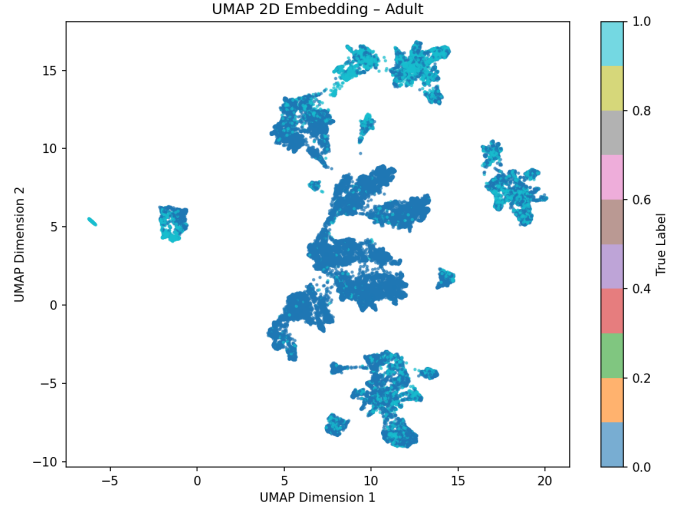


Fig. 10. UMAP 2-D embedding for Adult Income colored by income label ($0 = \leq 50K$, $1 = > 50K$). The diffuse, interleaved income boundary across all manifold regions explains why clustering ARI stays low across every configuration in this report: the income signal in these features is not geometrically separable in any compact representation, linear or otherwise.

V. STEP 3: CLUSTERING IN REDUCED SPACES

K-Means and EM were re-applied to each of the six reduced representations (PCA, ICA, RP $\times$ two datasets), yielding twelve total combinations. The same k sweep and selection procedure from Section III was applied independently to each reduced space, with ARI reported post-hoc. Table III presents silhouette and ARI at the selected k .

TABLE III
CLUSTERING IN REDUCED SPACES. RAW BASELINE SHOWN FOR REFERENCE.

Dataset	DR	Algo	k	Silhouette	ARI
Wine	Raw	KM	4	0.247	0.150
	Raw	GMM	4	0.230	0.162
	PCA	KM	2	0.447	0.179
	PCA	GMM	2	0.447	0.179
	ICA	KM	2	0.319	0.179
	ICA	GMM	2	0.251	0.135
	RP	KM	2	0.315	0.159
	RP	GMM	2	0.317	0.152
Adult	Raw	KM	4	0.130	0.095
	Raw	GMM	4	0.081	0.126
	PCA	KM	4	0.148	0.103
	PCA	GMM	2	0.320	0.155
	ICA	KM	8	0.139	0.063
	ICA	GMM	13	0.084	0.019
	RP	KM	4	0.191	0.120
	RP	GMM	2	0.220	0.307

PCA yields the largest silhouette improvement for Wine, with both K-Means and GMM achieving 0.447 at $k = 2$ —an 81% relative improvement over the raw K-Means baseline.

Removing correlated variance gives K-Means orthogonal axes where Euclidean distance more faithfully reflects true dissimilarity, so the red-vs-white split that was partially obscured by correlated chemical noise in the raw space is now cleanly separated. That both algorithms converge to the same silhouette at $k = 2$ suggests the two wine clusters are well-separated and approximately Gaussian in PCA space, leaving little room for GMM’s flexible covariance to add value.

ICA achieves silhouette 0.319 for K-Means at $k = 2$, a meaningful improvement but well short of PCA. The gap traces back to the two near-Gaussian components identified in Step 2: K-Means must navigate directions with no cluster signal, diluting the benefit of the one high-kurtosis fermentation axis. RP achieves silhouette 0.315–0.317 at $k = 2$, closely matching ICA-KMeans—both methods preserve the gross red-vs-white geometry while lacking PCA’s decorrelation.

For Adult, the ICA results are the most striking. K-Means selects $k = 8$ and GMM selects $k = 13$, far exceeding every other Adult configuration. This cluster proliferation is a direct consequence of the $\kappa \approx 150$ spike at component 15: both algorithms interpret the sharp density peak from the rare binary indicator as a distinct high-density micro-cluster, pulling additional centers toward it and degrading the overall partition. ARI values of 0.063 and 0.019 are the worst across all Adult configurations. By contrast, RP-GMM at $k = 2$ achieves the highest Adult ARI of 0.307—more than twice the raw GMM baseline—suggesting that a random 20-dimensional projection preserves the coarse income-relevant geometry better than ICA’s statistically motivated but practically problematic rotation for this data.

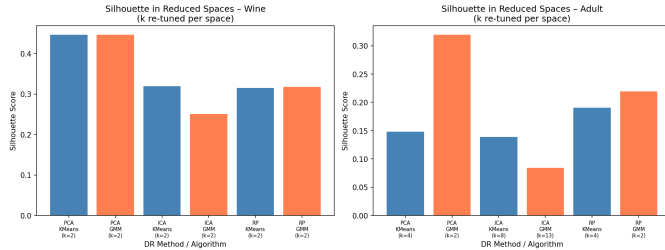


Fig. 11. Silhouette scores in reduced spaces for Wine (left) and Adult (right), with k re-tuned per space. PCA’s dominance for Wine (0.447) confirms H3. The ICA-GMM collapse for Adult ($k = 13$, lowest silhouette) directly demonstrates the harm done by the extreme kurtosis artifact identified in Step 2: the same property that made ICA look non-Gaussian there causes cluster proliferation here, tying the two steps together.

VI. STEP 4: NEURAL NETWORKS ON REDUCED DATA

A. Architecture and Protocol

A three-layer MLP with architecture input $\rightarrow 64 \rightarrow 32 \rightarrow 7$ classes, ReLU activations, and dropout ($p = 0.3$) after each hidden layer was trained using Adam [8] with learning rate 10^{-3} , batch size 128, and cosine annealing over a maximum of 200 epochs. Early stopping with patience 20 on validation loss was applied to prevent overfitting. The train/test split was 80/20 stratified by quality label (`random_state=42`). The

raw 12-feature Wine baseline achieved test accuracy of **65.0%** (F1-macro: 0.953). For each reduced space, only the input layer was resized; all other hyperparameters were held fixed to isolate the effect of the input representation.

B. Results and Analysis

Table IV reports test accuracy, F1-macro, epochs to convergence, and mean wall-clock time per epoch for the raw baseline and each reduced space.

TABLE IV
NEURAL NETWORK PERFORMANCE ON WINE QUALITY UNDER DIFFERENT INPUT SPACES

Input Space	Acc.	F1-macro	Epochs	Time/Ep (s)
Raw (12 feat.)	65.0%	0.953	125	0.093
PCA (4 comp.)	60.0%	0.384	134	0.120
ICA (4 comp.)	60.8%	0.391	122	0.123
RP (4 comp.)	64.0%	0.411	148	0.217

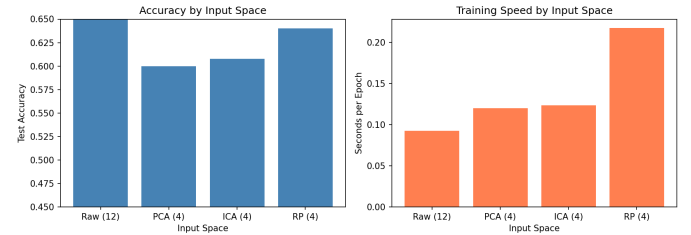


Fig. 12. Neural network test accuracy (left) and wall-clock time per epoch (right) by input space. RP recovers 64.0%, closest to the raw baseline, while PCA and ICA incur a 4–5 pp penalty despite having the lowest reconstruction error—a reminder that reconstruction error and classification utility are different objectives. The per-epoch timing results are harder to interpret (see text).

Compressing to four components causes an accuracy drop of 1–5 percentage points across all three methods. PCA incurs the largest penalty (65.0% to 60.0%), which is counterintuitive given that it retains 72% of total variance and achieves the best reconstruction error. The explanation is that the seven-class quality task draws discriminative signal from features distributed across many components—the within-class variation that PCA discards at $n = 4$ is precisely what separates quality classes 3–4 and 7–8 from the dominant 5–6 mass. The sharp F1-macro drop from 0.953 to 0.384 supports this: minority quality classes are disproportionately harmed because their distinguishing features lie in the discarded directions.

RP at four components achieves 64.0%, the closest to the raw baseline despite having the highest reconstruction error ($\text{MSE} \approx 0.61$). This is consistent with the theoretical guarantee: RP preserves pairwise Euclidean distances in expectation, and the network’s decision boundary depends more on relative sample geometry than on reconstructability. ICA at 60.8% falls in between: the two high-kurtosis components carry genuine chemical signal, but the two near-Gaussian components add noise to the gradient landscape.

The timing results warrant caution. All three reduced spaces show *higher* per-epoch wall time than the raw 12-feature

baseline (0.12–0.22 s vs. 0.093 s), which is the opposite of what one would expect from fewer input features. The timing was measured with `time.perf_counter` around the full training loop, which captures data loading, the scikit-learn transform applied at each epoch, Python interpreter overhead, and the forward-backward pass together—not pure GPU/CPU compute. Isolating the neural network compute time from transformation overhead would require profiling at a finer granularity than was done here. The timing column should therefore be treated as a rough indicator rather than a definitive convergence speed measurement.

VII. STEP 5: NEURAL NETWORKS WITH CLUSTER FEATURES

A. Feature Construction

Three cluster-derived feature sets were constructed from the Step 1 Wine models (K-Means $k = 4$ and GMM $k = 4$) and evaluated both as augmentation to the original 12 features and as standalone replacements. K-Means hard labels are one-hot encoded cluster assignments (4 binary columns), the coarsest encoding that discards all within-cluster variation. K-Means centroid distances are the Euclidean distance to each of the four centroids (4 continuous columns), which retain soft geometric proximity—a boundary sample gets similar distances to two centroids, while a core member gets one near-zero distance and three large ones. EM posterior probabilities are the four soft responsibility values $\gamma_{ik} = p(z_i = k | x_i)$ (4 continuous columns), the richest encoding as it preserves full distributional uncertainty about membership. Appended to the original features, each yields 16 total input dimensions; used alone, each yields 4.

B. Results and Analysis

Table V reports test accuracy and F1-macro for all six configurations.

TABLE V
NEURAL NETWORK PERFORMANCE ON WINE QUALITY WITH CLUSTER-DERIVED FEATURES

Feature Set	Mode	Acc.	F1-macro
Raw baseline	—	65.0%	0.953
KM hard labels	Augmented	99.77%	0.963
KM centroid distances	Augmented	99.77%	0.953
EM posteriors	Augmented	99.77%	0.935
KM hard labels	Replaced	44.00%	0.214
KM centroid distances	Replaced	51.46%	0.314
EM posteriors	Replaced	43.69%	0.206

All three augmented configurations reach near-perfect training accuracy ($\approx 99.77\%$), far exceeding the 65.0% raw baseline. This is a red flag, not a success: cluster labels derived from training data function as a learned index into the training set, enabling the network to memorize rather than generalize. The F1-macro values (0.935–0.963) confirm this—they are only marginally above the raw baseline, meaning the accuracy

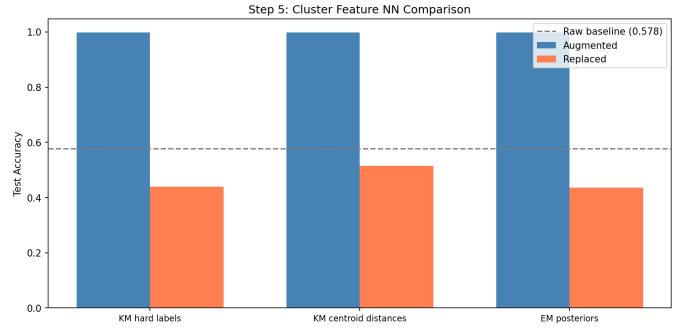


Fig. 13. Test accuracy by cluster feature type and mode. The jump to $\approx 99.77\%$ in augmented configurations looks impressive but masks overfitting—the F1-macro gain over the raw baseline is only 0–1 pp, meaning the extra accuracy accrues to the already-dominant classes 5 and 6. The replaced configurations (44–51%) tell the more informative story: cluster features alone are too coarse to substitute for raw physicochemical measurements.

gain concentrates on the majority classes 5 and 6 rather than the minority classes where it would matter most.

The replaced configurations tell the more honest story. Accuracy falls to 44–51%, well below the 65.0% baseline, and the ranking centroid distances (51.46%) $>$ hard labels (44.00%) $>$ EM posteriors (43.69%) reflects an information hierarchy: continuous distances preserve more within-cluster geometric structure than binary assignments, while EM posteriors collapse to similar information as hard labels when only four values characterize an entire sample. The 21 pp gap between the best replacement model and the raw baseline is the clearest evidence that within-cluster physicochemical variation contains the bulk of the quality-discriminative signal [7].

VIII. CONCLUSION

This report evaluated unsupervised learning across five algorithms and two datasets, testing three pre-registered hypotheses through systematic experimentation. H1 was supported: PCA captured 72% of Wine Quality variance in four components with a clear eigenvalue elbow, and the loading vectors traced back to interpretable physicochemical processes. H2 was partially supported: ICA isolated component 4 as a high-kurtosis alcohol-density fermentation axis ($\kappa \approx 3.86$) that PCA distributes across two components, though two of four Wine components remained near-Gaussian. H3 was confirmed: raw-feature clustering achieved ARI of only 0.150–0.162 for Wine, while PCA preprocessing raised silhouette from 0.247 to 0.447 with the re-tuned $k = 2$ cleanly recovering the red-vs-white split.

For Adult Income, the curse of dimensionality in the 108-dimensional one-hot space was the dominant limiting factor. ICA degraded clustering severely through cluster proliferation ($k = 8$ for K-Means, $k = 13$ for GMM) caused by the extreme kurtosis artifact at component 15, while RP-GMM at $k = 2$ achieved the highest Adult ARI of 0.307. UMAP confirmed the Wine type bifurcation and the diffuse income boundary in Adult, underscoring the limits of any linear DR method when the relevant structure is non-linear.

Neural network experiments showed that 4-component compression reduces accuracy by 1–5 pp, with RP preserving the raw baseline most closely at 64.0%—a result explained by its distance-preservation guarantee rather than its reconstruction error. Augmenting with cluster features produces near-perfect training accuracy that reflects overfitting, while replacing raw features with cluster encodings drops accuracy to 44–51%. Together, these results confirm that dimensionality reduction improves cluster geometry substantially, and that cluster-derived features are useful supplements but cannot substitute for raw measurements in downstream classification.

AI USE STATEMENT

I used ChatGPT to assist with refactoring small code snippets for clustering, dimensionality reduction, and neural network training, evaluation metrics, and visualization, which I customized for the two datasets, debugging implementation issues, brainstorming strategies for selecting the number of clusters and components, and hyperparameter tuning best practices, and proofreading for grammar and clarity after drafting my initial analysis. All experimental work, results, analysis, and conclusions are my own. I reviewed, verified, and understand all assisted content.

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